



Registration No. :

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Total number of printed pages – 3

B. Tech.  
Subject Code: MA203

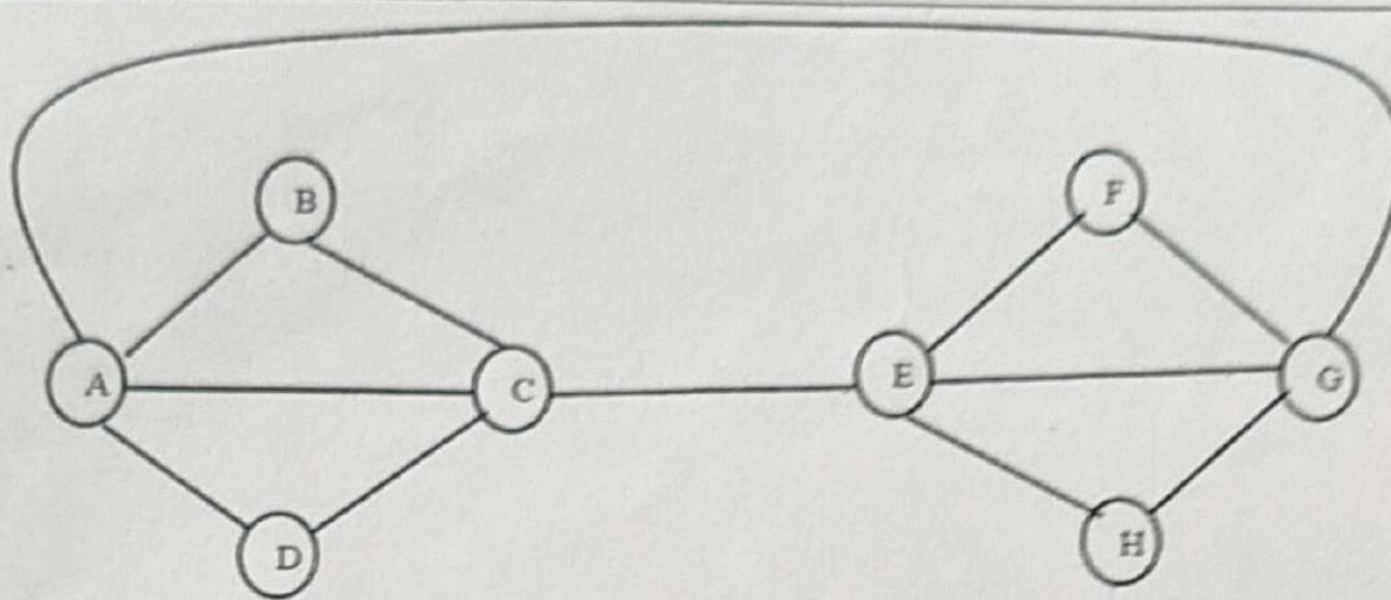
3<sup>RD</sup> SEMESTER REGULAR EXAMINATION – 2024  
DISCRETE MATHEMATICS AND GRAPH THEORY  
BRANCH: CSE AND ALL ALLIED BRANCHES

Time – 3 Hours

Full Marks – 100

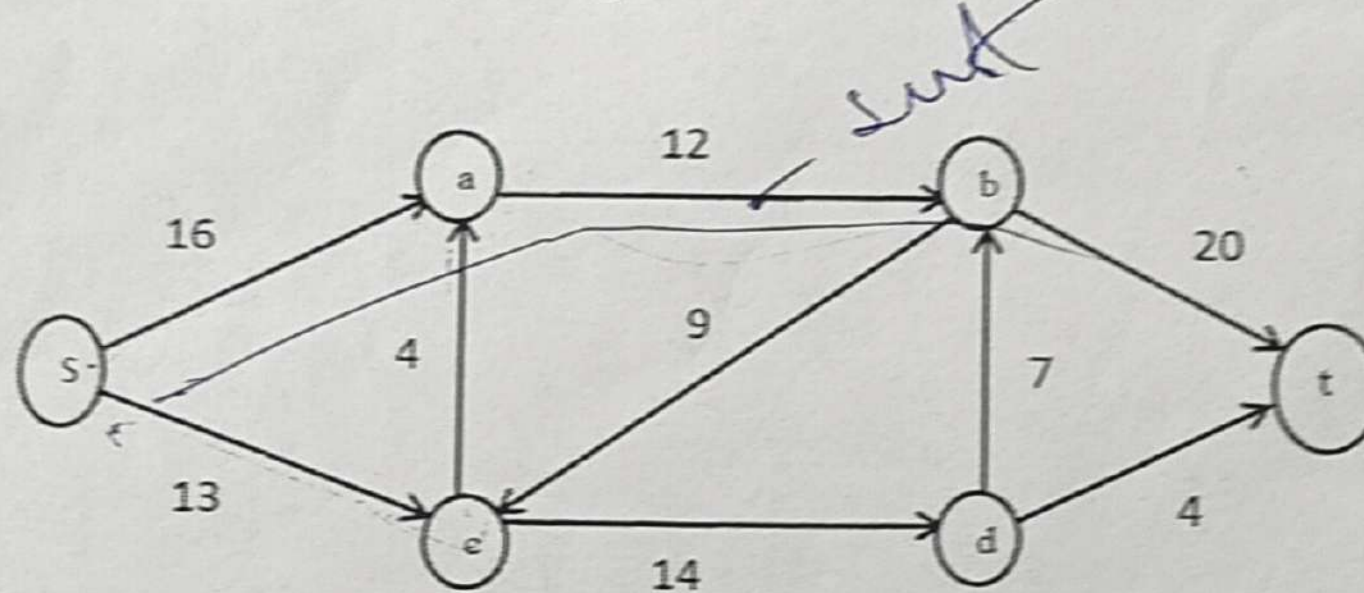
Answer all Questions from Part – A and Part – B

| Q1. | Part-A (ANSWER ALL QUESTIONS)                                                                                                             | CO  | Marks | BL |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|----|
| (a) | Show that $P \vee \sim P$ is a tautology.                                                                                                 | CO1 | 2     | 2  |
| (b) | State De Morgan's law for quantifier.                                                                                                     | CO1 | 2     | 1  |
| (c) | Find the maximal and minimal elements of the partially ordered set $(\{1,2,3,4,6,8\},/)$                                                  | CO1 | 2     | 2  |
| (d) | How many solutions does the equation $x + y + z = 17$ have, where $x, y, z$ are non-negative integers ?                                   | CO2 | 2     | 1  |
| (e) | Find the generating function for the sequence 0,1,2,3,4....                                                                               | CO2 | 2     | 2  |
| (f) | Evaluate the particular solution of the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 7^n$ for all $n \geq 2$ .                        | CO2 | 2     | 2  |
| (g) | If 6 colours are used to paint 37 home. Show that at least 7 home of them will be of same colour.                                         | CO2 | 2     | 2  |
| (h) | A graph contain 21 edges, 3 vertices of degree 4 and all other vertices of degree 2. Calculate the total number of vertices in the graph. | CO3 | 2     | 1  |
| (i) | How many edges does $K_{2,6}$ contain?                                                                                                    | CO3 | 2     | 1  |
| (j) | Find the chromatic number of complete graph $K_5$ .                                                                                       | CO3 | 2     | 2  |
| (k) | Define complete binary tree.                                                                                                              | CO4 | 2     | 1  |
| (l) | Find all the spanning trees of complete graph $K_4$ .                                                                                     | CO4 | 2     | 2  |

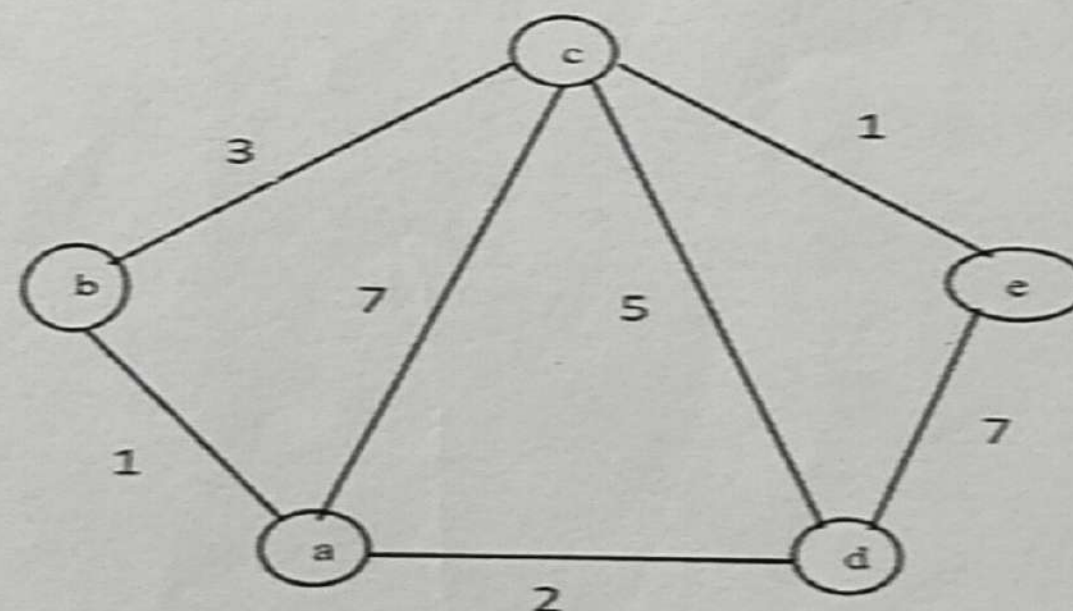


Q5. Answer the followings:

- (a) Find the maximum flow in the transport network in the following figure and determine the corresponding minimum cut.



- (b) Find the shortest distance and shortest path from 'a' to 'e' in the given weighted graph using Dijkstra algorithm.



Q6. Answer the followings:

- (a) Show that the set of all positive rational numbers forms an abelian group under the composition define by  $a * b = \frac{ab}{2}$

- (b) Determine the  $(2, 5)$  group code function  $e_H : B^2 \rightarrow B^5$  for the parity

matrix  $H = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$



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Total number of printed pages – 3

B.Tech

Subject Code: MA108

3<sup>RD</sup> SEMESTER REGULAR EXAMINATION – 2025

DISCRETE MATHEMATICS

BRANCH: CSE and all Allied Branches

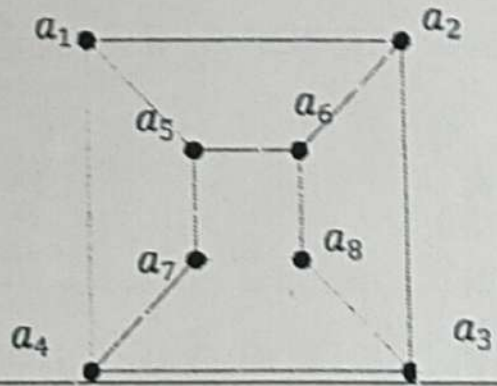
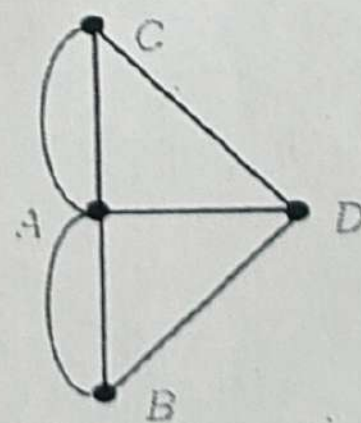
Time – 3 Hours

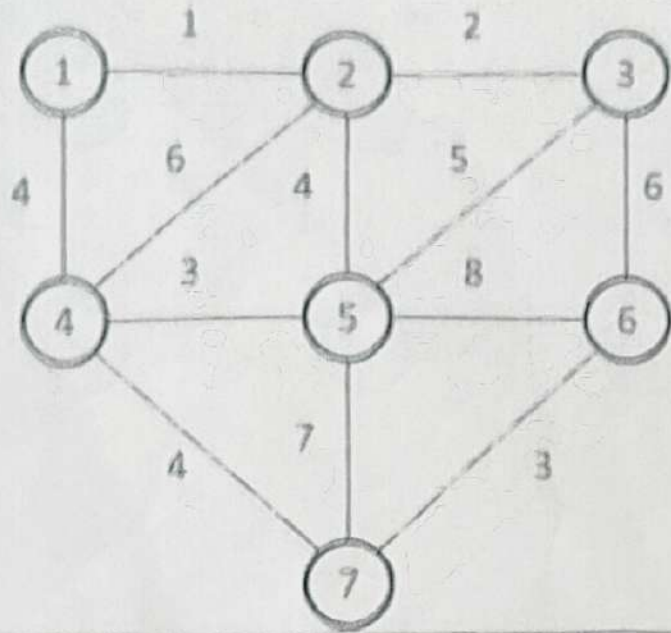
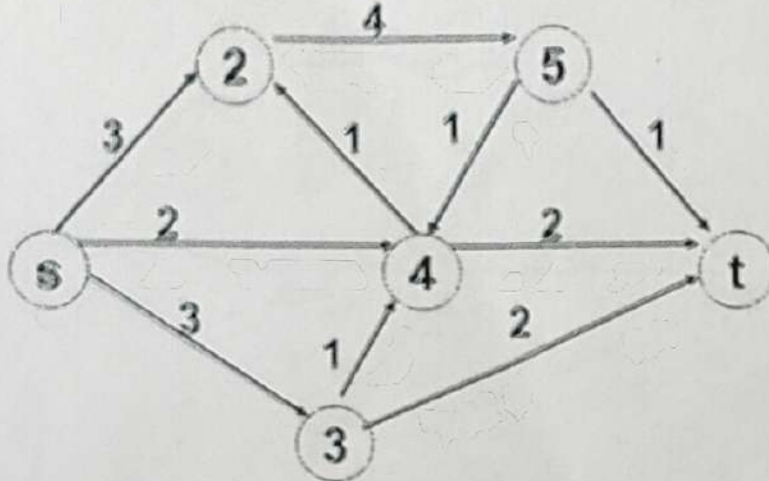
Full Marks – 100

Answer all Questions from Part – A and Part – B

| Q1. | Part-A (ANSWER ALL QUESTIONS)                                                                                                           | CO  | Marks | BL |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------|-----|-------|----|
| (a) | Construct the truth table for the relation $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$ .                            | CO1 | 2     | 3  |
| (b) | Use mathematical induction to prove that $n^2 + n$ is divisible by 2 for all $n \in \mathbb{N}$ .                                       | CO1 | 2     | 2  |
| (c) | Find the transitive closure of the relation $R = \{(1, 2), (2, 3)\}$ on the set $\{1, 2, 3\}$                                           | CO1 | 2     | 3  |
| (d) | How many solutions are there of $x + y + z = 17$ subject to the constraints $x \geq 1, y \geq 2$ and $z \geq 3$ .                       | CO2 | 2     | 3  |
| (e) | Solve the recurrence relation $a_n = 3a_{n-1} + 2$ with $a_0 = 2$ .                                                                     | CO2 | 2     | 2  |
| (f) | How many ways can 8 people be arranged in a line if two specific people must stand together?                                            | CO2 | 2     | 3  |
| (g) | State Handshaking theorem.                                                                                                              | CO3 | 2     | 1  |
| (h) | Write the Euler's formula for a planer graph with 4 components.                                                                         | CO3 | 2     | 1  |
| (i) | What is the chromatic number of an odd cycle?                                                                                           | CO3 | 2     | 1  |
| (j) | Define spanning tree. Give an example.                                                                                                  | CO4 | 2     | 1  |
| (k) | A tree has two vertices of degree 2, one vertex of degree 3 and three vertices of degree 4. How many vertices of degree 1 does it have? | CO4 | 2     | 1  |
| (l) | State the Max-Flow Min-Cut Theorem for a network.                                                                                       | CO4 | 2     | 1  |
| (m) | Show that the set of integers does not form a group under multiplication.                                                               | CO5 | 2     | 3  |

|     |                                                              |     |   |   |
|-----|--------------------------------------------------------------|-----|---|---|
| (n) | Define a subgroup of a group. Give an example.               | CO5 | 2 | 1 |
| (o) | Construct a binary code with a minimum Hamming distance of 2 | CO5 | 2 | 3 |

|            | Part-B                                                                                                                                                                                                                                                                   | CO  | Marks | BL |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|----|
| <b>Q2.</b> | <b>Answer any two:</b>                                                                                                                                                                                                                                                   |     |       |    |
| (a)        | Prove using a truth table that $(P \vee Q) \wedge (\neg P \vee R)$ is logically equivalent to $(P \wedge R) \vee (Q \wedge \neg R)$ .                                                                                                                                    | CO1 | 7     | 2  |
| (b)        | Apply Warshall's Algorithm to compute the transitive closure of a relation represented by the adjacency matrix below: $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$                                                   | CO1 | 7     | 2  |
| (c)        | Define a poset. Draw the Hasse diagram for the divisors of 60 under the "divides" relation and verify whether that forms a poset or not.                                                                                                                                 | CO1 | 7     | 1  |
| <b>Q3.</b> | <b>Answer any two:</b>                                                                                                                                                                                                                                                   |     |       |    |
| (a)        | Use generating function to solve the recurrence relation $a_n = 3a_{n-1} + 2$ with $a_0 = 1$                                                                                                                                                                             | CO2 | 7     | 2  |
| (b)        | Solve $a_{n+2} - 4a_{n+1} + 4a_n = 2^n$                                                                                                                                                                                                                                  | CO2 | 7     | 3  |
| (c)        | In how many ways can 5 people be seated in a row if two of them, Alice and Bob, must sit next to each other?                                                                                                                                                             | CO2 | 7     | 2  |
| <b>Q4.</b> | <b>Answer any two:</b>                                                                                                                                                                                                                                                   |     |       |    |
| (a)        | Define bipartite Graph. Check whether the following graph is bipartite or not?                                                                                                       | CO3 | 7     | 1  |
| (b)        | Define Euler circuit. State the necessary and sufficient condition for a connected graph to have a Euler circuit.<br>Check whether the following graph has a Euler circuit or not.  | CO3 | 7     | 1  |
| (c)        | State and prove Euler's formula for a connected planer simple graph                                                                                                                                                                                                      | CO3 | 7     |    |

|     |                                                                                                                                                                                                  |     |   |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---|
| Q5. | Answer the followings:                                                                                                                                                                           |     |   |   |
| (a) | <p>Use Kruskal's algorithm find a minimal spanning tree (MST ) of the following graph.</p>                     | CO4 | 7 | 2 |
| (b) | <p>Find the maximal flow of the following network using Ford-Fulkerson algorithm</p>                          | CO4 | 7 | 3 |
| Q6. | Answer the followings:                                                                                                                                                                           |     |   |   |
| (a) | Prove that the fourth roots of unity $1, -1, i, -i$ form an abelian multiplicative group                                                                                                         | CO5 | 7 | 3 |
| (b) | For the $(2,4)$ encoding function determine how many errors can detect and how many errors can correct with respect to following data $e(00) = 0000, e(10) = 0110, e(01) = 1011, e(11) = 1100$ . | CO5 | 7 | 3 |

Best of Luck



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Total number of printed pages -04

B.Tech.  
MTH21105

3<sup>rd</sup> SEMESTER BACK EXAMINATION – 2025

SUBJECT NAME: Discrete Mathematics

Branch(s): CSIT

Time – 3 Hours

Full Marks – 100

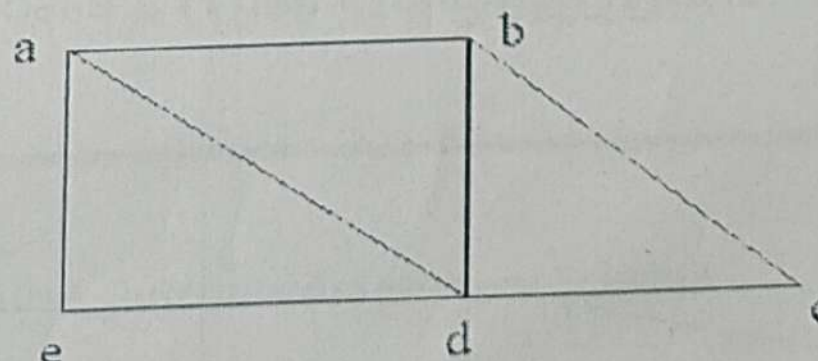
Answer all Questions from Part – A and Part – B  
The figures in the right hand margin indicate marks.

**PART – A**

1. Answer the following questions:

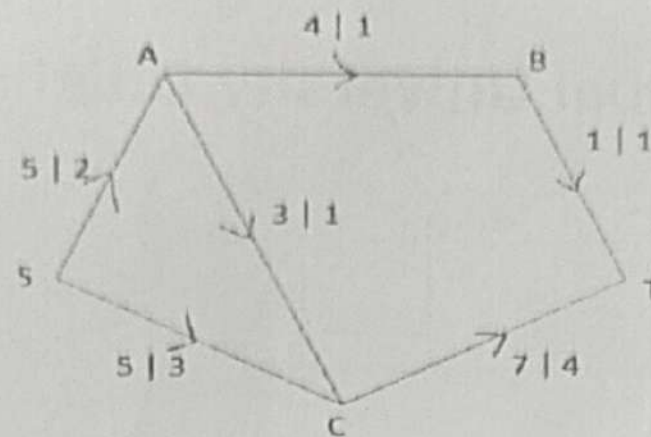
[15x 2]

- Show that  $q \rightarrow (i \vee p \wedge q)$  is a tautology.
- Find the maximal and minimal elements of the partially ordered set  $(\{2,4,5,10,12,20,25\}, |)$ .
- Let  $n$  be an integer and  $3n+2$  be odd. Then show that  $n$  is odd by method of contrapositive.
- Find number of derangements of the set  $\{1,2,3,4\}$ .
- Find the particular solution of the recurrence relation  $a_n - 7a_{n-1} + 10a_{n-2} = 4^n$ , for all  $n \geq 2$ .
- State Pigeonhole principle.
- Show that  $K_5$  is not planar.
- How many edges does  $K_{3,6}$  contain?
- Determine whether the following graph has Euler circuit or Euler path and find it.



(j) Define spanning tree.

(k) Check whether the chain SCABT flow augmented path or not?



- (l) Find the generating function for the sequence  $1, 1, 1, 1, \dots$ .
- (m) Find the distance between  $x = 110110$ , and  $y = 010010$ .
- (n) Write all the subgroups of the group  $G = (\mathbb{Z}_6, +_6)$ .
- (o) Find all the generators of the cyclic group  $G = (\{1, 3, 7, 9\}, \times_{10})$ .

## PART - B

2. Answer any two:

[7 x 2]

- (a) Show that the division relation on set of Integer is a partial ordering relation. Draw the Hasse diagram representing the partial ordering  $\{(a, b) | a \text{ divides } b\}$  on  $\{1, 2, 3, 4, 6, 8, 12\}$ .
- (b) Find the transitive closure of the relation  $R = \{(1, 2), (2, 1), (2, 3), (3, 4)\}$  on the set  $A = \{1, 2, 3, 4\}$  using Warshall's algorithm.
- (c) Prove that  $n^3 - n$  is divisible by 3 for every positive integer  $n$  by using mathematical induction.

3. Answer any two:

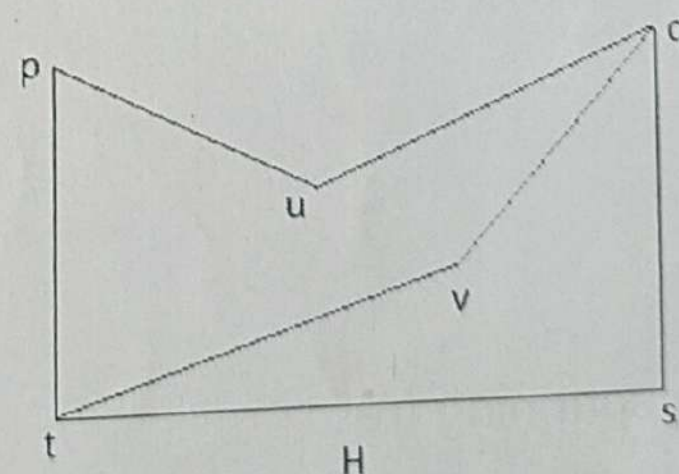
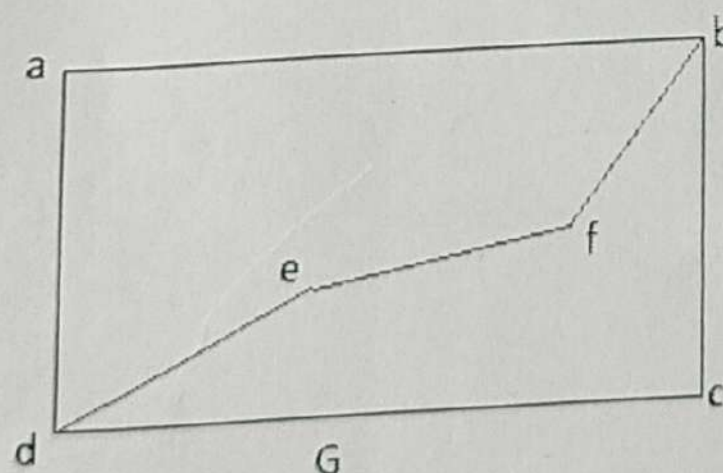
[7 x 2]

- (a) Solve the recurrence relation  $a_n - 6a_{n-1} + 9a_{n-2} = 0$ , for all  $n \geq 2$  by using generating function.
- (b) How many integers between 1 and 500 (inclusive) are
  - (i) divisible by at least one of 3, 5, 7?
  - (ii) divisible by 3 and 5 but not by 7?
- (c) Find the sequence  $b_r$  from the generating function  $\sum_{r=0}^{\infty} b_r x^r = \frac{1}{x^2 - 5x + 6}$

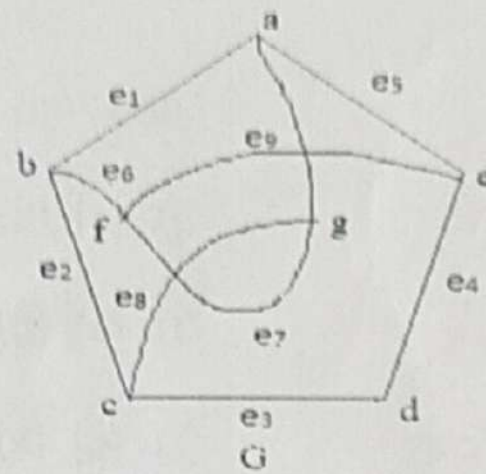
4. Answer any two:

[7 x 2]

- (a) Prove that the number of vertices with odd degree is always even in a simple graph.
- (b) Test whether the following graphs G and H are isomorphic or not.



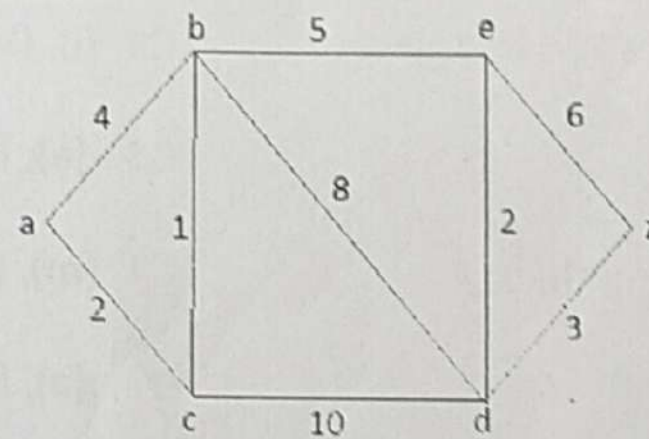
(c) Define adjacency matrix and incidence matrix and also find the adjacency and incidence matrices of the following graph G.



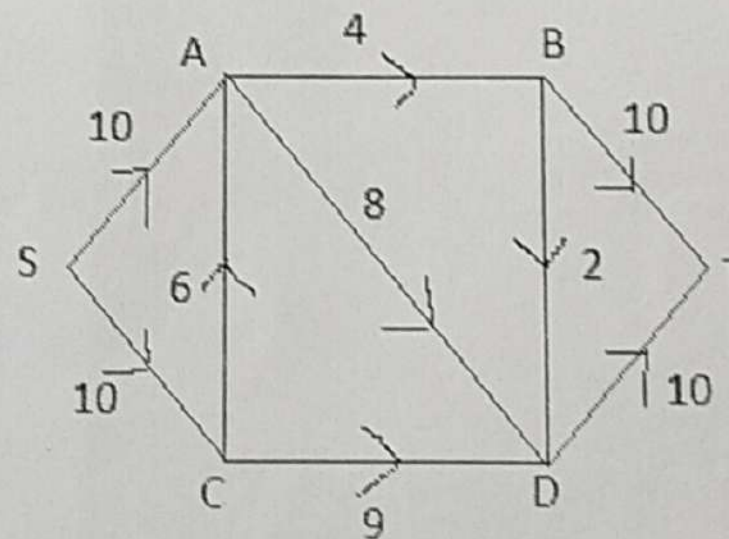
5. Answer the following:

[7 x 2]

(a) Find the length of a shortest path between the vertices a and z by using Dijkstra's algorithm in the following graph



(b) Find the maximum flow of the following network using Ford-Fulkerson algorithm.



6. Answer the following:

[7 x 2]

(a) Check whether the algebraic structure  $G = (\left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in Q \right\}; *)$ , where  $A * B = A.B$  forms a group or not?

(b) Show that (2,5) encoding function defined by  $e(00)=00000$ ,  $e(01)=01110$ ,  $e(10)=10101$ ,  $e(11)=11011$ , is a group code.



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Total number of printed pages – 3

B.Tech  
Subject Code: MA203

3<sup>RD</sup> SEMESTER SUMMER QUARTER EXAMINATION – 2025  
SUBJECT NAME: DISCRETE MATHEMATICS AND GRAPH THEORY

BRANCH: CSE and all Allied Branches

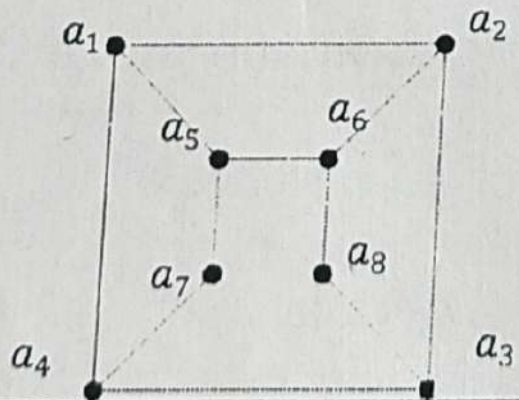
Time – 3 Hours

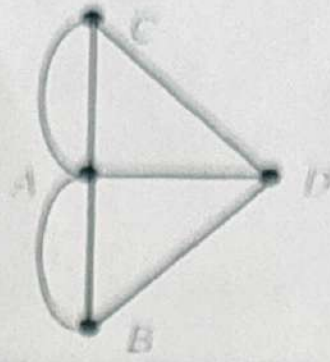
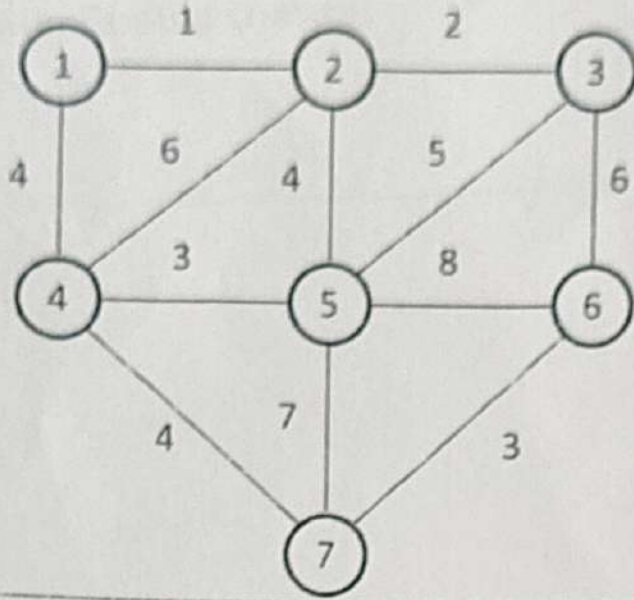
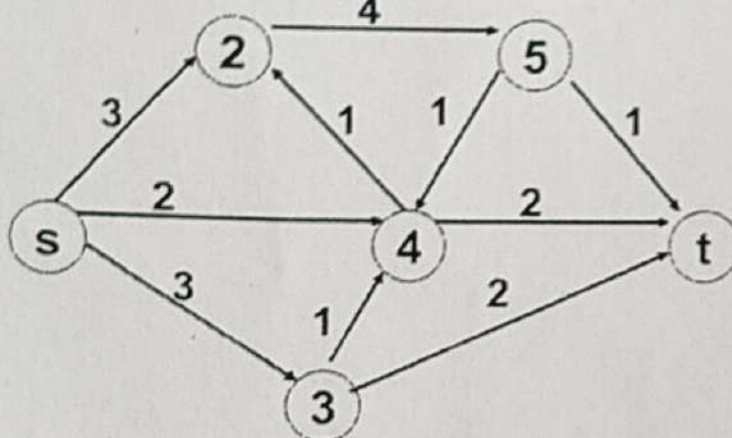
Full Marks – 100

Answer all Questions from Part – A and Part – B

| Part-A (ANSWER ALL QUESTIONS) |                                                                                                                                            | CO  | Marks | BL |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|----|
| Q1.                           |                                                                                                                                            |     |       |    |
| (a)                           | Construct the truth table for the relation $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$ .                               | CO1 | 2     | 3  |
| (b)                           | Use mathematical induction to prove that $n^2 + n$ is divisible by 2 for all $n \in \mathbb{N}$ .                                          | CO1 | 2     | 2  |
| (c)                           | Find the transitive closure of the relation $R = \{(1, 2), (2, 3)\}$ on the set $\{1, 2, 3\}$                                              | CO1 | 2     | 3  |
| (d)                           | What is the maximum number of groups into which you can distribute 30 students such that at least one group must have at least 5 students? | CO2 | 2     |    |
| (e)                           | Solve the recurrence relation $a_n = 3a_{n-1} + 2$ with $a_0 = 2$ .                                                                        | CO2 | 2     |    |
| (f)                           | How many ways can 8 people be arranged in a line if two specific people must stand together?                                               | CO2 | 2     |    |
| (g)                           | State Handshaking theorem.                                                                                                                 | CO3 | 2     |    |
| (h)                           | Write the Euler's formula for a planer graph with 4 components.                                                                            | CO3 | 2     |    |
| (i)                           | What is the chromatic number of an odd cycle?                                                                                              | CO4 | 2     |    |
| (j)                           | Define spanning tree. Give an example.                                                                                                     | CO4 | 2     |    |
| (k)                           | If T is a binary of height 8, then what will be the maximum number of terminal vertex for T?                                               | CO4 | 2     |    |
| (l)                           | State the Max-Flow Min-Cut Theorem for a network.                                                                                          |     |       |    |

|     |                                                                           |     |   |   |
|-----|---------------------------------------------------------------------------|-----|---|---|
| (m) | Show that the set of integers does not form a group under multiplication. | CO5 | 2 | 3 |
| (n) | Define a subgroup of a group. Give an example.                            | CO5 | 2 | 1 |
| (o) | Construct a binary code with a minimum Hamming distance of 2              | CO5 | 2 | 3 |

|            | Part-B                                                                                                                                                                                                                 | CO  | Marks | BL |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|----|
| <b>Q2.</b> | <b>Answer any two:</b>                                                                                                                                                                                                 |     |       |    |
| (a)        | Prove using a truth table that $(P \vee Q) \wedge (\neg P \vee R)$ is logically equivalent to $(P \wedge R) \vee (Q \wedge \neg R)$ .                                                                                  | CO1 | 7     | 2  |
| (b)        | Apply Warshall's Algorithm to compute the transitive closure of a relation represented by the adjacency matrix below: $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$ | CO1 | 7     | 2  |
| (c)        | Define a poset. Draw the Hasse diagram for the divisors of 60 under the "divides" relation and verify whether that forms a poset or not.                                                                               | CO1 | 7     | 1  |
| <b>Q3.</b> | <b>Answer any two:</b>                                                                                                                                                                                                 |     |       |    |
| (a)        | Use generating function to solve the recurrence relation $a_r - 3a_{r-1} + 2a_{r-2} = 0$ for $r=2,3,4,\dots$ and initial condition $a_0 = 2$ , and $a_1 = 3$                                                           | CO2 | 7     | 2  |
| (b)        | What is the solution of the recurrence relation $a_n = a_{n-1} + a_{n-2} + 3n + 1 \text{ for } n \geq 2,$ with $a_0 = 2$ and $a_1 = 3$ ?                                                                               | CO2 | 7     | 3  |
| (c)        | In how many ways can 5 people be seated in a row if two of them, Alice and Bob, must sit next to each other?                                                                                                           | CO2 | 7     |    |
| <b>Q4.</b> | <b>Answer any two:</b>                                                                                                                                                                                                 |     |       |    |
| (a)        | Define bipartite Graph. Check whether the following graph is bipartite or not?                                                     | CO3 | 7     | 1  |
| (b)        | Define Euler circuit. State the necessary and sufficient condition for a connected graph to have a Euler circuit.                                                                                                      | CO3 | 7     | 1  |

|     |                                                                                                                                                                                                         | Marks | Total Mark |   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------|---|
|     | <p>Check whether the following graph has a Euler circuit or not.</p>                                                  |       |            |   |
| (c) | Prove that every planar graph has a vertex of degree at most 5.                                                                                                                                         | CO3   | 7          | 2 |
| Q5. | Answer the followings:                                                                                                                                                                                  |       |            |   |
| (a) | <p>Use Kruskal's algorithm find a minimal spanning tree (MST ) of the following graph.</p>                           | CO4   | 7          | 2 |
| (b) | <p>Find the maximal flow of the following network using Ford-Filkerson algorithm</p>                                | CO4   | 7          | 3 |
| Q6. | Answer the followings:                                                                                                                                                                                  |       |            |   |
| (a) | <p>State Lagrange's Theorem.</p> <p>If <math>G</math> is a group of order 10 and <math>H</math> is a subgroup of order 5, how many distinct left cosets does <math>H</math> have in <math>G</math>?</p> | CO5   | 7          | 1 |
| (b) | Construct a binary linear code using a generator matrix and explain how syndrome decoding can be used to correct errors.                                                                                | CO5   | 7          | 3 |

**Best of Luck**